

HAOMING WANG

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RESEARCH INTERESTS

Spatial intelligence, Multimodal LLMs reasoning, generative AI on mobile devices.

EDUCATION

Sept 2022 – May 2027 (Anticipated) **Ph.D. in Electrical and Computer Engineering**
University of Pittsburgh, Pittsburgh, PA, USA
Advisor: Prof. Wei Gao, Intelligent System Lab

Sept 2018 – May 2022 **B.Eng. in Automation**
Zhejiang University, Hangzhou, China
Chu Kochen Honors College

GPA: 3.8/4.0

EXPERIENCE

Graduate Student Researcher

Intelligent System Lab, University of Pittsburgh, Pittsburgh, PA, USA

- **Spatial Reasoning Performance Improvement (Sep 2025 – Present).** Improved VLM spatial reasoning via representation-level training and inference-time pipelines: Dirichlet-energy regularizer shaping a VLM latent subspace that encodes 3D scene topology [5]; training-free multi-view fusion for small on-device VLMs, deployed on Jetson Orin, AR Glass, and smartphone [7]; VLM-based autonomous mobile systems for small-object search [10] and infrastructure-assisted navigation [11].
- **Spatial Reasoning Benchmarking and Analysis (May 2025 – Jan 2026).** Established benchmarks and diagnostic methodologies for VLM spatial reasoning: customizable framework generating photo-realistic 3D test scenes with user-controllable complexity axes [1]; cognitively grounded multi-view benchmark with reasoning-path / latent-state analysis [6]; survey of tasks, benchmarks, and methods [12].
- **Efficient Model Training on Mobile Devices (Sep 2022 – May 2025).** Designed efficient training systems for heterogeneous mobile / IoT fleets: explainable warm-start for on-device LLM personalization [2]; gradient-corrected federated AIoT training under data and device heterogeneity [3]; staleness-tolerant federated learning via gradient inversion [4].

PUBLICATIONS (PEER-REVIEWED)

1. **[CVPR'26 Oral] InfiniBench: Infinite Benchmarking for Visual Spatial Reasoning with Customizable Scene Complexity**
Haoming Wang, Qiyao Xue, Wei Gao.
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 (Acceptance 25.4%, Oral).
[\[arXiv\]](#) [\[Code\]](#) [\[Data\]](#)
2. **[MobiSys'25] Never Start from Scratch: Expediting On-Device LLM Personalization via Explainable Model Selection**
Haoming Wang, Boyuan Yang, Xiangyu Yin, Wei Gao.
Proceedings of the 23rd ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2025 (Acceptance 18.0%). [\[Paper\]](#)
3. **[MobiCom'25] When Device Delays Meet Data Heterogeneity in Federated AIoT Applications**
Haoming Wang, Wei Gao.
Proceedings of the 31st ACM International Conference on Mobile Computing and Networking (MobiCom), 2025 (Acceptance 17.1%). [\[Paper\]](#)

4. **[AAAI'25] Tackling Intertwined Data and Device Heterogeneities in Federated Learning with Unlimited Staleness**
Haoming Wang, Wei Gao.
Proceedings of the 39th AAAI Conference on Artificial Intelligence, 2025 (Acceptance 23.4%). [\[arXiv\]](#)

RECENT PREPRINTS / UNDER REVIEW (FIRST AUTHOR)

5. **Uncovering and Shaping the Latent Representation of 3D Scene Topology in Vision-Language Models**
Haoming Wang, Wei Gao.
Under Review, 2026. [\[Paper\]](#) [\[Code\]](#)
6. **Reasoning Path and Latent State Analysis for Multi-view Visual Spatial Reasoning: A Cognitive Science Perspective**
Qiyao Xue, Haoming Wang (co-first), Weichen Liu, Shiqi Wang, Yuyang Wu, Wei Gao.
Under Review, 2026. [\[arXiv\]](#)
7. **MosaicThinker: On-Device Visual Spatial Reasoning for Embodied AI via Iterative Construction of Space Representation**
Haoming Wang, Qiyao Xue, Weichen Liu, Wei Gao.
Under Review, 2026. [\[arXiv\]](#)
8. **Deciphering Personalization: Towards Fine-Grained Explainability in Natural Language for Personalized Image Generation Models (FineXL)**
Haoming Wang, Wei Gao.
Under Review, 2025. [\[Paper\]](#)

OTHERS

9. **FreezeAsGuard: Mitigating Illegal Adaptation of Diffusion Models via Selective Tensor Freezing**
Kai Huang, Haoming Wang (co-first), Wei Gao.
Preprint, 2024. [\[Paper\]](#) [\[Code\]](#)
10. **HEVEN: Small Object Search and Navigation on VLM-Based Autonomous Mobile Systems**
Anonymous Authors (incl. Haoming Wang, 3rd author).
Double-Blind Under Review, 2026.
11. **GlobalNav: Daily Object Navigation in VLM-Based Autonomous Mobile Systems with Aligned Local and Global Views**
Anonymous Authors (incl. Haoming Wang, 3rd author).
Double-Blind Under Review, 2026.
12. **Spatial Reasoning in Multimodal Large Language Models: A Survey of Tasks, Benchmarks and Methods**
Weichen Liu, Qiyao Xue, Haoming Wang, Xiangyu Yin, Boyuan Yang, Wei Gao.
Under Review, 2026.

HONORS & AWARDS

April 2026 **ECE Research Assistant of the Year**
Department of Electrical and Computer Engineering, University of Pittsburgh

TECHNICAL SKILLS

On-Device / Edge: Jetson Orin (PyTorch + TensorRT), Meta AR Glass, Android (ONNX / NNAPI) **Programming:** Python, PyTorch, C/C++.